

Ideation Phase

Define the Problem Statements

Date	JUNE 2026
Team ID	
Project Name	Plugging-into-the-Future-An-Exploration-of-Electricity-Consumption-Patterns
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	A data-driven analyst and energy enthusiast who explores electricity consumption patterns to uncover trends, identify peak demand, and support smarter, sustainable energy decisions.			
I'm trying to	Analyze electricity consumption data from various regions and time periods to identify usage trends, peak demand periods, seasonal variations, and opportunities for energy efficiency and sustainable planning.			
But	The data is large and complex, with many variables and missing values, making it challenging to find meaningful insights and spot patterns across different timeframes and locations.			
Because	There are no intuitive tools or dashboards that simplify this electricity dataset into actionable visual insights — most users are not data analysts and need easier access.			
Which makes me feel	Overwhelmed, but motivated — because understanding electricity usage can lead to smarter decisions, better energy management, and a more sustainable future.			

I am	I'm trying to	But	Because	Which makes me feel
A data analyst focused on exploring electricity consumption patterns to support smarter energy management and sustainable development.	Understand electricity usage trends, identify peak demand periods, and analyze seasonal patterns across regions to support efficient energy planning.	The electricity data is large, complex, and contains missing values, which makes it difficult to uncover meaningful insights.	There are limited visualization tools and intuitive dashboards that can easily turn raw electricity data into clear, actionable insights.	Motivated to find insights that can improve energy management and contribute to a more sustainable future, despite the challenges.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Energy Analyst / Sustainability Officer	Analyze electricity consumption data and uncover usage trends to support energy efficiency and sustainability initiatives.	The raw data is large, complex, and lacks contextual features such as time, location, and customer segments.	The dataset includes multiple columns with inconsistent formats (dates, numerical values, categorical attributes), making analysis and visualization challenging across different dimensions and time periods.	Overwhelmed and uncertain, as it's hard to extract meaningful insights without proper data preparation and exploration.
PS-2	Energy Researcher / Data Analyst	Identify electricity consumption patterns by region, time, and season to find peak demand periods and support data-driven decisions for energy management.	I can't easily find meaningful patterns because the data is unstructured and lacks clear categorization for time, location, and usage types.	Essential insights are hidden within large datasets that need cleaning, aggregation, and visualization to reveal trends, correlations, and anomalies.	Frustrated and challenged, as it takes more time and effort to prepare the data before I can uncover valuable insights.